that the information originated from the signer and has not been altered by a cracker in the middle. Digital signatures can provide the added assurances of evidence to the origin, identity and status, as well as acknowledging the consent of the sender.

Digital signatures use a standard, worldwide accepted format, called Public Key Infrastructure (PKI), to provide the highest levels of security and universal acceptance. In many countries, digital signatures have the same legal significance as the traditional forms of signed documents. Digital signatures are widely used for avoiding forging or tampering of important documents such as financial documents or credit card data.



A security token is a hardware component that are used to identify and authenticate users.



17.4.3 Authentication protocols

At present, there are two kinds of security authentication protocols widely used in E-Commerce, namely Secure Electronic Transaction (SET) and Secure Sockets Layer (SSL).

Secure Electronic Transaction

Secure Electronic Transaction (SET) is a security protocol for electronic payments with credit cards, in particular via the Internet. SET was developed in 1996 by VISA and MasterCard, with the participation of GTE, IBM, Microsoft and Netscape.

The implementation of SET is based on the use of digital signatures and the encryption of transmitted data with asymmetric and symmetric encryption algorithms. SET also use dual signatures to ensure the privacy.

The SET purchase involves three major participants: the customer, the seller and the payment gateway. Here the customer shares the order information with the seller but not with the payment gateway. Also the customer shares the payment information only with the payment gateway but not with the seller. So, with the SET, the credit card number may not be known to the seller and will not be stored in seller's files also could not be recovered by a hacker.

The SET protocol guarantees the security of online shopping using credit cards on the open network. It has the advantages of ensuring the integrity of transaction data and the non-repudiation of transactions. Therefore, it has become the internationally recognized standard for credit card online transaction.

SET system incorporates the following key features:

- Using public key encryption and private key encryption ensure data confidentiality.
- Use information digest technology to ensure the integrity of information.
- Dual signature technology to ensure the identity of both parties in the transaction.

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